Lab 5: Multiple Linear Regression

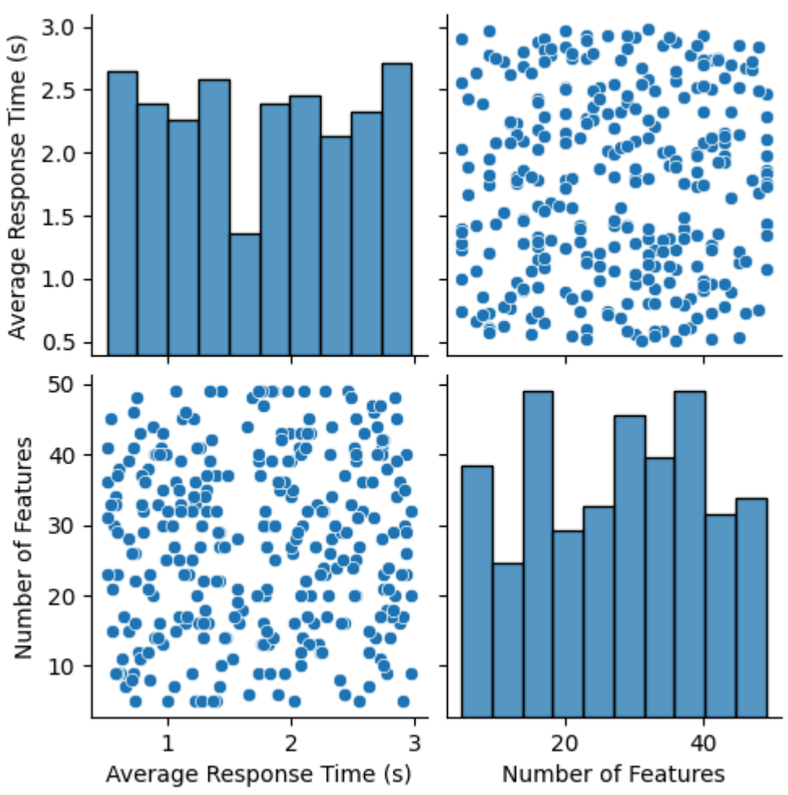
Step-by-Step Instructions:

In [1]:
```
# Hello everyone! In this lab we'll walk through a multiple linear regression example using synthetic mobile application data.

import numpy as np  # Importing numpy for numerical operations
import pandas as pd  # Importing pandas for data manipulation
import matplotlib.pyplot as plt  # Importing matplotlib for plotting
from sklearn.linear_model import LinearRegression  # Importing LinearRegression from sklearn
from sklearn.metrics import mean_squared_error  # Importing mean_squared_error from sklearn
from sklearn.model_selection import train_test_split  # Importing train_test_split from sklearn
import seaborn as sns  # Importing seaborn for advanced visualizations

# Let's start by loading our dataset into a pandas DataFrame.
raw_dataset = pd.read_csv('Synthetic_app_data.csv')  # Reading the CSV file into a DataFrame

# Displaying the first 5 rows of the dataset to get an idea of what it looks like.
print(raw_dataset.head())  # Displaying the first 5 rows of the DataFrame
```

```
   Average Response Time (s)  Number of Features  Number of Bugs Reported  \
0                   1.436350                  49                       16
1                   2.876786                  36                       12
2                   2.329985                  34                        0
3                   1.996646                  39                        1
4                   0.890047                  44                        8

   Training Hours Provided  User Satisfaction Score
0                 3.970126                89.980978
1                 3.100364                56.732146
2                 2.667305               121.502856
3                 4.469463               124.535638
4                 3.942986               129.077397
```

In [2]:
```
# Creating a pairplot to visualize the relationships between different variables.
sns.pairplot(raw_dataset[['Average Response Time (s)', 'Number of Features']])  # Pairplot to visualize the relationship between 'Average Response Time (s)' and 'Number of Features' remem
```

Out[2]: <seaborn.axisgrid.PairGrid at 0x7fea6e358a30>



In [3]:
```
# Printing the column names of the dataset.
print(raw_dataset.columns)  # Displaying the column names of the DataFrame

# Separating the independent variables (features) and dependent variable (target).
indp_vars = raw_dataset[['Average Response Time (s)', 'Number of Features', 'Number of Bugs Reported', 'Training Hours Provided']]  # Selecting multiple columns as independent variables
dep_var = raw_dataset['User Satisfaction Score']  # Selecting 'User Satisfaction Score' as the dependent variable

# Splitting the data into training and testing sets.
train_x, test_x, train_y, test_y = train_test_split(indp_vars, dep_var, test_size=0.2, random_state=42)  # Splitting the data with 80% for training and 20% for testing
```

```
Index(['Average Response Time (s)', 'Number of Features',
       'Number of Bugs Reported', 'Training Hours Provided',
       'User Satisfaction Score'],
      dtype='object')
```

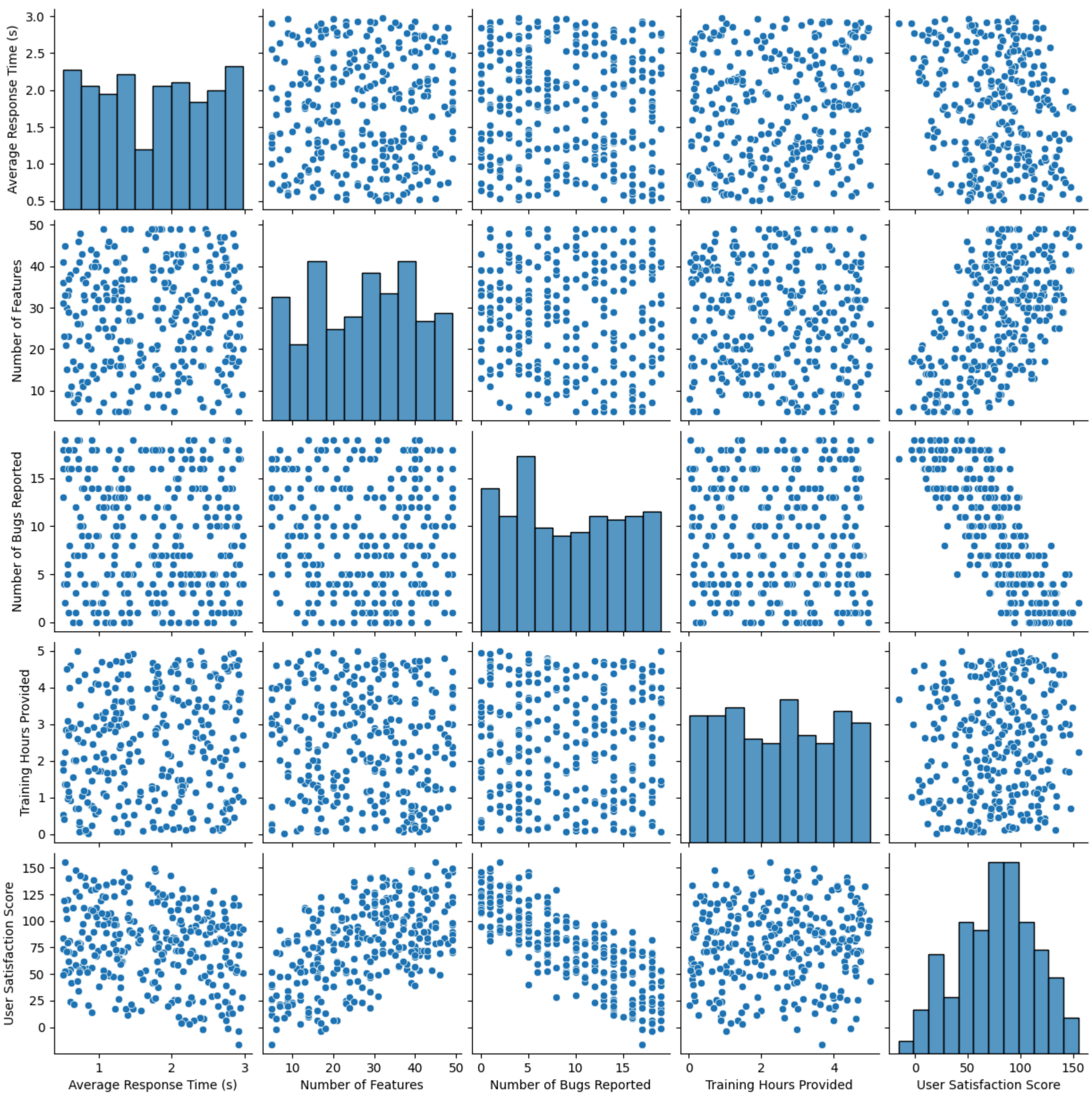
In [4]:
```
# Creating a Linear Regression model and fitting it to the training data.
lr_model = LinearRegression()  # Creating an instance of LinearRegression
lr_model.fit(train_x, train_y)  # Fitting the model to the training data

# Using the model to predict User Satisfaction Score on the test data.
pred = lr_model.predict(test_x)  # Making predictions on the test data

# Calculating and printing the Mean Squared Error of the model's predictions.
mse = mean_squared_error(pred, test_y)  # Calculating the Mean Squared Error between predicted and actual values
print("Mean Squared Error:", mse)  # Printing the Mean Squared Error
```

```
Mean Squared Error: 21.341088894194325
```

Part 1: MLR

In [9]:
```
#Step 1/2
raw_dataset
sns.pairplot(raw_dataset)
```

Out[9]: <seaborn.axisgrid.PairGrid at 0x7fea4fd4a760>



Observations:

- **Avg Response Time vs User Satisfaction** -> Negative correlation higher response time lower satisfaction
- **Number of Features vs User Satisfaction** -> Slight positive trend more features may improve satisfaction
- **Number of Bugs vs User Satisfaction** -> Strong negative correlation more bugs lower satisfaction
- **Training Hours vs User Satisfaction** -> Moderate positive correlation more training slightly improves satisfaction
- **Outliers** -> Some extreme values like high response time with moderate satisfaction worth investigating
- **Key Predictors** -> Response time and bugs have the strongest impact on satisfaction

In [10]:
```
#Step 4
X = raw_dataset[['Average Response Time (s)', 'Number of Features', 'Number of Bugs Reported', 'Training Hours Provided']]
y = raw_dataset['User Satisfaction Score']

# Split dataset 80% train, 20% test
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print(f"Training Data: {X_train.shape}, Testing Data: {X_test.shape}")
```

```
Training Data: (240, 4), Testing Data: (60, 4)
```

In [11]:
```
#step 5
model = LinearRegression()
model.fit(X_train, y_train)

print("Model Coefficients:", model.coef_)
print("Intercept:", model.intercept_)
```

```
Model Coefficients: [-14.96954945   1.50200003  -4.92889966   3.0268404 ]
Intercept: 99.04540366638571
```

In [16]:
```
#Step 6
from sklearn.metrics import r2_score

y_pred = model.predict(X_test)

mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print(f"Mean Squared Error: {mse}")
print(f"R2 Score: {r2}")
```

```
Mean Squared Error: 21.341088894194325
R2 Score: 0.9835002165140093
```

Part 2: : MLR vs LR

In [15]:
```
X_single = raw_dataset[['Average Response Time (s)']]
y = raw_dataset['User Satisfaction Score']

# Split dataset 80% train, 20% test
X_train_single, X_test_single, y_train, y_test = train_test_split(X_single, y, test_size=0.2, random_state=42)


slr_model = LinearRegression()
slr_model.fit(X_train_single, y_train)
```

```
y_pred_slr = slr_model.predict(X_test_single)

mse_slr = mean_squared_error(y_test, y_pred_slr)
r2_slr = r2_score(y_test, y_pred_slr)

mse_slr, r2_slr
```

Out[15]: (1166.1542277019535, 0.0983921971484133)

MLR vs SLR Comparison

MLR

MSE 21.34 R2 0.9835

SLR using avg response time

MSE 1166.15 R2 0.0984

Observations

MLR way better lower MSE higher R2 explains 98 percent of variance

SLR bad high MSE low R2 only explains 9 percent

Response time alone bad predictor

MLR considers all factors bugs features training hours better accuracy

Conclusion

User satisfaction depends on multiple factors not just response time

MLR better model more accurate predictions

In [ ]: